INCH-POUND
MIL-DTL-23053/13B
17 June 1996
SUPERSEDING
MIL-I-23053/13A
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, FLUOROELASTOMER, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

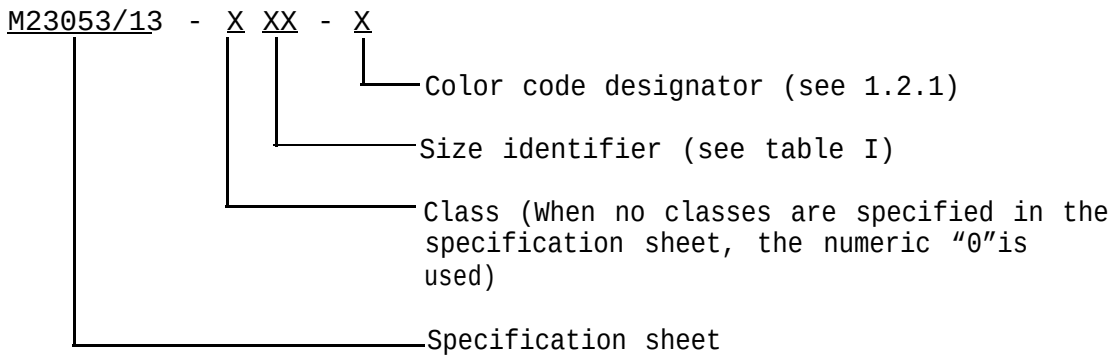
Polymer type: The base elastomer used in formulating this sleeving shall be fluoroelastomer.

Continuous operating temperature range: -40°C (-40°F) to +200°C (+392°F).

Color: The heat shrinkable sleeving shall be furnished in the color specified by the acquisition activity (see 6.2b); however, black shall be considered standard. Colors shall conform to the requirements of Class II of MIL-STD-104.

Longitudinal change: ± 20 percent.

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



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Example: Black, 0.500 inch (12.7 mm) as supplied ID sleeving shall be identified as M23053/13-O04-0.

TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/13-001-*	.187(4.7)	.094(2.4)	.035 ± .007(.9 ± .18)
M23053/13-O02-*	.250(6.4)	.125(3.2)	.035 ± .007(.9 ± .18)
M23053/13-O03-*	.375(9.5)	.187(4.7)	.035 ± .007(.9 ± .18)
M23053/13-O04-*	.500(12.7)	.250(6.4)	.035 ± .007(.9 ± .18)
M23053/13-O05-*	.625(15.9)	.312(7.9)	.042 ± .008(1.1 ± .20)
M23053/13-O06-*	.750(19.1)	.375(9.5)	.042 ± .008(1.1 ± .20)
M23053/13-O07-*	.875(22.2)	.437(11.1)	.049 ± .011(1.2 ± .28)
M23053/13-O08-*	1.000(25.4)	.500(12.7)	.049 ± .011(1.2 ± .28)
M23053/13-O09-*	1.250(31.8)	.625(15.9)	.055 ± .015(1.4 ± .38)
M23053/13-010-*	1.500(38.1)	.750(19.1)	.055 ± .015(l.4 ± .38)
M23053/13-011-*	2.000(56.8)	1.000(25.4)	.065 ± .017(1.7 ± .43)
M23053/13-012-*	.125(3.21	.063(1.6)	.065 ± .017(l.7 ± .43)

1/ Diameter limits for object to be enclosed shall be as specified in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The asterisk in the part number shall be replaced by color code designations (see 1.2.1). Black is the standard color.

Unrestricted shrinkage: Test method 4.6.5; 200° ± 5°C (392° ± 9°F) for 10 minutes maximum.

Table II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Cold impact	No cracking	4.6.7.2 -40° ± 1° (-40° ± 2°)

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TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
Heat shock	No cracking, flowing or dripping	4.6.8 300° ± 5°C (572° ± 9°F)
Restricted shrinkage	No cracks	4.6.6.1.2 200° ± 2°C (392° ± 4°F)
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile stength, psi (MPa), minimum	1,200 (8.2)	4.6.13, ASTM D412
Ultimate elongation, percent,	250	4.6.12, ASTM D412
Tensile stress at 200 percent elongation, psi (MPa), maximum	2,000(13.8)	4.6.2 ASTM D412
Dielectric strength, volts/roil, (Kv/mm), minimum	200(7.9)	4.6.2 ASTM D2671
Volume resistivity, Ohm-cm, minimum	10^{9}	4.6.2 ASTM D876
Corrosion	No corrosion or pitting	4.6.10.1 and 4.6.10.2 175° ± 2°(347° ± 4°F) for 16 hours.

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Heat resistance, properties after:		4.6.9 250° ± 3°C (482° ± 5.4°F) for 168 hours
Tensile strength, psi (MPa), miminum	1,200(8.2)	
Ultimate elongation, percent, minimum	200	
Water absorption, percent, maximum	0.5	4.6.2 ASTM D570, 24 hrs at 23°C
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	1,200 (8.2)	
Ultimate elongation, percent, minimum	250	
Flammability	Self-extinquishing within 15 seconds	4.6.14 Procedure A ASTM D2671
Fungus resistance	No growth	4.6.2 ASTM G21

Storage life: The supplier shall certify storage at 18° to 35°C (65° to 95°F) for a period of 2 years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: Heat shrinkable, flexible, fluoroelastomer sleeving is used in high temperature applications or where exposure to high temperature solvents is expected.

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Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - MI
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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